# Chhaya

A Type 2 diabetes digital twin that shows its work.

One Sensor Wear | Scored on Held-Out Patients | Misses Published Openly

SX

# Team SynapseX

## D.B.S. College, Kanpur

Happiest Health Digital Twin Challenge 2026 · Submission SynapseX_D.B.S Kanpur

# Nine Bars Written Before the Runs

Pre-registered evaluation protocol across Gates 1, 2, and 3.

**5 PASSED**

Gate 2 Reveal & Fingersticks

**4 MISSED**

Gate 3 Excursions & Record Prior

# Consequences in the Clinical Product

GATE 3 MISSED

## No Meal Alert

No excursion probability or alert shown anywhere in the interface.

RECORD PRIOR MISSED

## Fusion Added Nil

Health record retained strictly as background context for clinician review.

PLAIN BASELINE WON

## Plain Average Used

Converted fingerstick average drives patient summary in clinical practice.

HARDWARE ECONOMICS

# ₹ 4,400 to 5,100

## The cost of a single 14-day continuous glucose sensor in India.

A patient wears one once and is treated from that report for months.

# When the Sensor Comes Off

The clinic has a static paper report, occasional fingersticks, and a meal log.

Static Paper PDF | Sparse Fingersticks | Patient Meal Diary

# What the Clinic Needs to Know

01 **How long does the report stay true?**

02 **Which readings should you not believe?**

03 **What keeps the shadow true?**

04 **When does the report go stale?**

# The Reveal Protocol

STEP 01

## 1. Calibrate

Fit seven parameters on k days of sensor wear.

STEP 02

## 2. Hide

Conceal subsequent continuous sensor data.

STEP 03

## 3. Estimate

Project shadow using meals and fingersticks.

STEP 04

## 4. Reveal

Overlay ground truth sensor to score error.

# Open Datasets & Licenses

SHANGHAIT2DM · CC BY 4.0

## 100 Patients, 109 Wears

Inpatient care: 15-min sensor, fingersticks, diet text, drugs, laboratory panels.

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## 45 Free-Living People

Libre + Dexcom sensors, Fitbit wristband, and meal macronutrients.

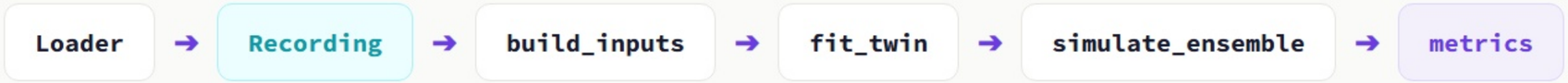
# Split by Patient, Never by Row

`chhaya.config.is_dev_patient()`

Priors and parameters tuned on dev patients.
Held-out test patients evaluated strictly once.

# One-Way Data Architecture

The browser computes nothing: every curve is precomputed in static JSON bundles.



# Personalised Physiological Model

EXTENDED DES IN JAX (X64)

## ODE System at 1-Min Steps

Two-compartment gut absorption, circadian dawn sinusoid, and beta-cell response.

BAYESIAN ESTIMATION

## MAP Mode + Laplace Band

L-BFGS-B mode fitting with Hessian curvature giving the 80% uncertainty envelope.

# Estimate = 50% Physiology + 50% Average Day

Errors are only partly correlated: the blend beats both individual models.

PHYSIOLOGY (50%)

## Metabolic Dynamics

Knows exact meal grams and timings logged today.

AVERAGE DAY (50%)

## Empirical Habits

Knows persistent routine habits the meal log missed.

BARS P1, P2, P3 · PASSED

# 1.4 mg/dL Closer

**6% gain over the patient's own average day (p = 4e-05, 19 held-out patients).**

Advantage does not fade over 7 unmeasured days (0.4 to 2.0 mg/dL).

# Fingersticks Evaluation

BAR F1 · PASSED

## 2.8 mg/dL Closer Live

Paired RMSE difference, 95% CI 1.6 to 4.7 mg/dL, 86% of patients, p = 4e-06.

BAR F2 · PASSED

## 9.5 mg/dL In Hindsight

Paired difference with 95% interval 4.6 to 12.1 mg/dL on held-out patients.

BAR P1 · MISSED

# 0.02 mg/dL

## Giving the twin fasting glucose as prior changed error by only 0.02 mg/dL (p = 0.16).

The health record is not worth any sensor days. Fusion is built and measured: it is not a gain.

# Sensor History Alone Beat the Twin

| RECORD ONLY | SENSOR HISTORY | FINGERSTICKS | FUSED TWIN |
|---|---|---|---|
| 0.566 | 0.626 | 0.642 | 0.596 |
| AUPRC | AUPRC (Winner) | AUPRC | AUPRC (Missed) |

Calibration slope 0.482 vs target 0.8–1.2. Result: no meal alerts in the product.

# Empirical Truth & Baselines

REPORT LEVEL AVERAGE

## 6.4 vs 8.9 mg/dL

Plain converted fingerstick average was closer than our twin in hindsight. Plain average is used.

STALENESS DRIFT PROMPT

## AUROC 0.82 vs 0.82

Cumulative surprise sum tied plain fingerstick average vs report (difference 0.00).

# Explicit Scientific Boundaries

01 **No Indian patient dataset (Shanghai & US cohorts).**

02 **Supervised inpatient care data.**

03 **Hidden unmeasured windows at most 11 days.**

04 **Physiology model advantage is ~2% over baseline.**

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# Chhaya

**chhaya-digital-twin.vercel.app**

Code, tests, and decision records public on GitHub · Team SynapseX · D.B.S Kanpur